

Astronomy

(Scheduled Event - 30 Minutes)



Description: Participants will be assessed on their understanding of the formation, classification, evolution, and life cycles of stars.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Teams must bring: Pen/Pencil

Teams may not bring:

- Notes or guides of any kind.

Optional - teams may bring:

- Four-function calculator

The Competition: This event may be administered as a written test or as a series of stations which may include diagrams, photographs, spectra, H-R diagrams, data analysis, models, and problems for students to solve.

Content topics will include but are not limited to:

1. Star Formation and Properties

- Describe how stars form from nebulae through protostar and main-sequence stages.
- Explain the roles of gravity, pressure, and nuclear fusion in star formation.
- Compare stars based on mass, size, temperature, luminosity, and color.

2. Classification and Stellar Evolution

- Classify stars using the Hertzsprung-Russell (H-R) Diagram and basic spectral classes.
- Identify and compare main-sequence stars, giants, supergiants, white dwarfs, neutron stars, and black holes.
- Describe the evolutionary pathways of low-mass stars (red giant → planetary nebula → white dwarf) and high-mass stars (red supergiant → supernova → neutron star or black hole).
- Explain how stellar mass, color, and temperature influence a star's classification, lifespan, and evolution.

3. The Sun

- Identify the major layers of the Sun and their functions.
- Explain how the Sun produces energy.
- Describe the Sun's current stage in the stellar life cycle.

4. Representative Objects

Students should be familiar with the significance and classification of:

- The Sun
- Orion Nebula
- Proxima Centauri
- Sirius
- Betelgeuse
- Crab Nebula

5. Data Analysis and Applications

- Interpret simplified diagrams, spectra, H-R diagrams, graphs, tables, and astronomical images.
- Apply concepts of star formation, classification, and the basic stages of stellar evolution to analyze data and solve problems.

Examples:

- Given an H-R diagram, identify which star is hottest.
- Compare the brightness of two stars.
- Determine which star is likely to become a red giant.
- Match star colors with approximate temperatures.
- Identify the stage of a star's life from a diagram.
- Interpret a simple spectrum to determine whether two stars contain the same element.
- Use a graph of stellar brightness over time to identify a variable star.

Scoring:

- Points will be awarded for each correct response

Tiebreakers:

- Ties will be broken by the accuracy or quality of answers to select questions chosen by the event leader prior to the competition.

Resources:

- [Stars](#)
- [Stellar Evolution](#)
- [Life of a Star](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles